ReAltoR: Real Estate Search Engine

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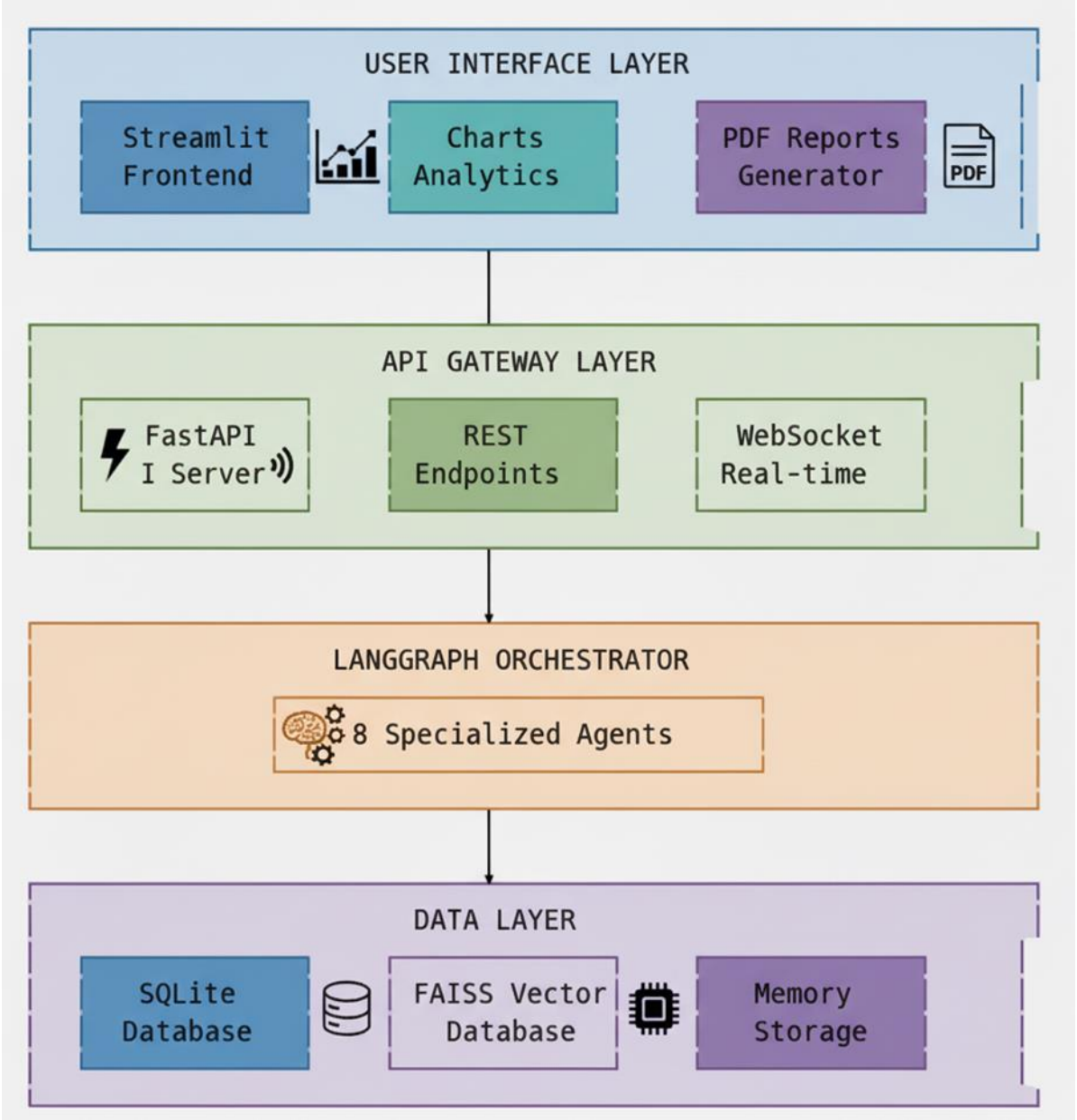
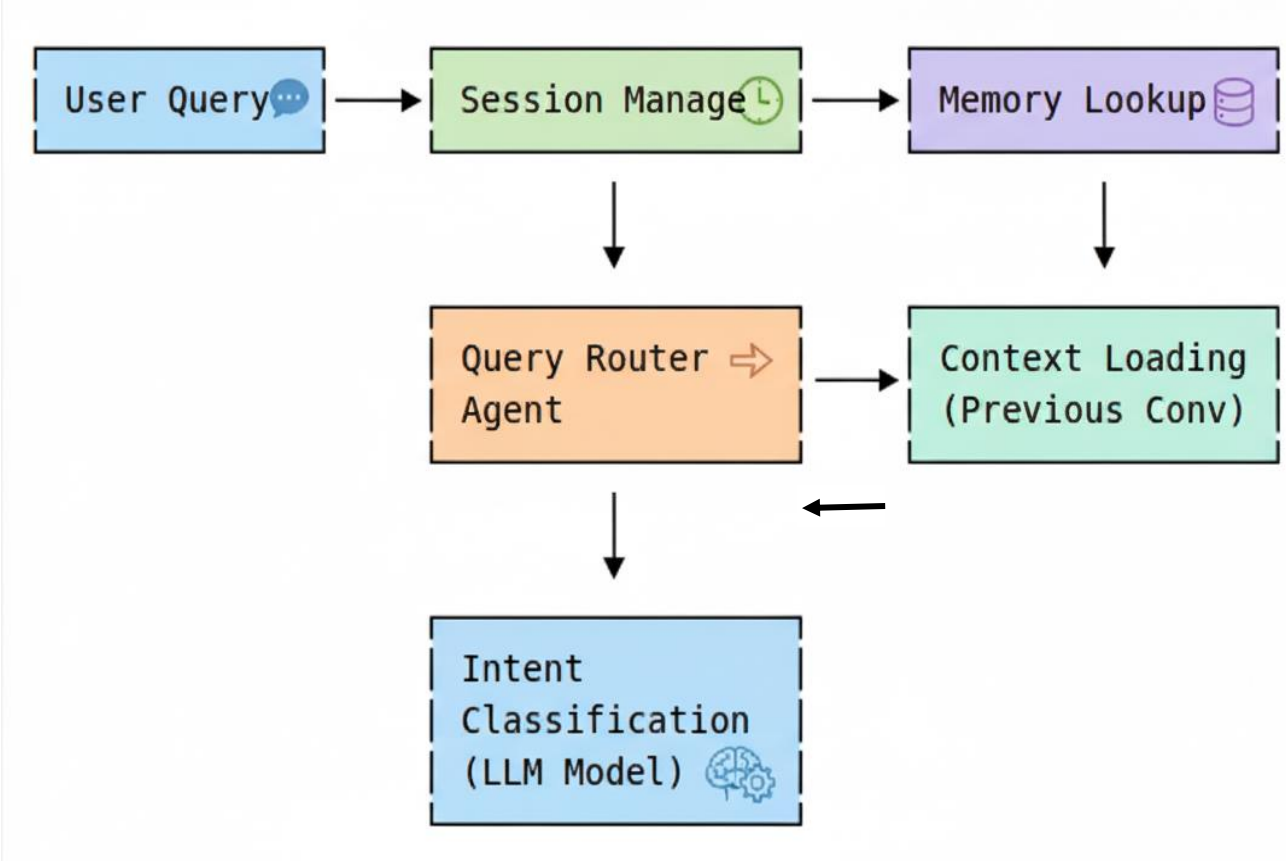
System Architecture

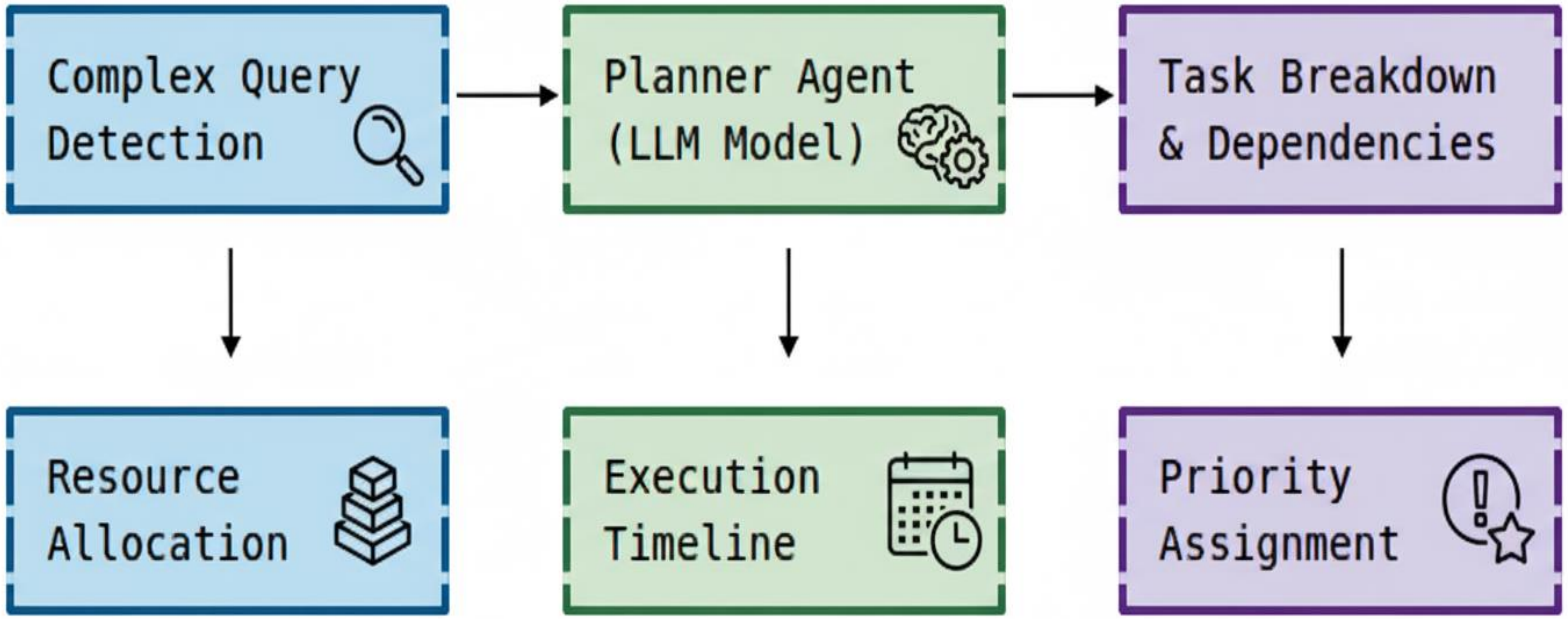


Phase 1



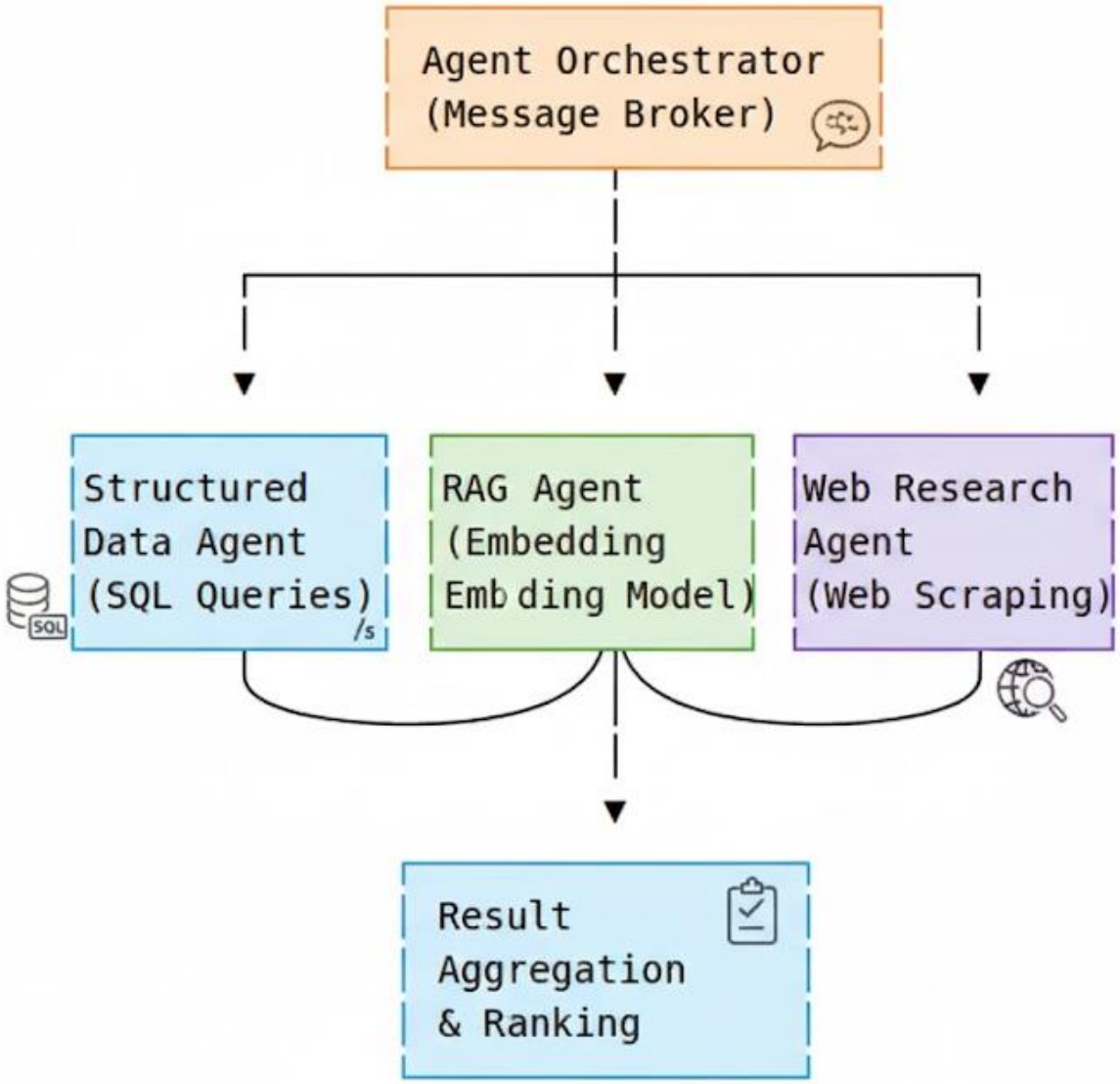
Query Processing and Routing

Phase 2



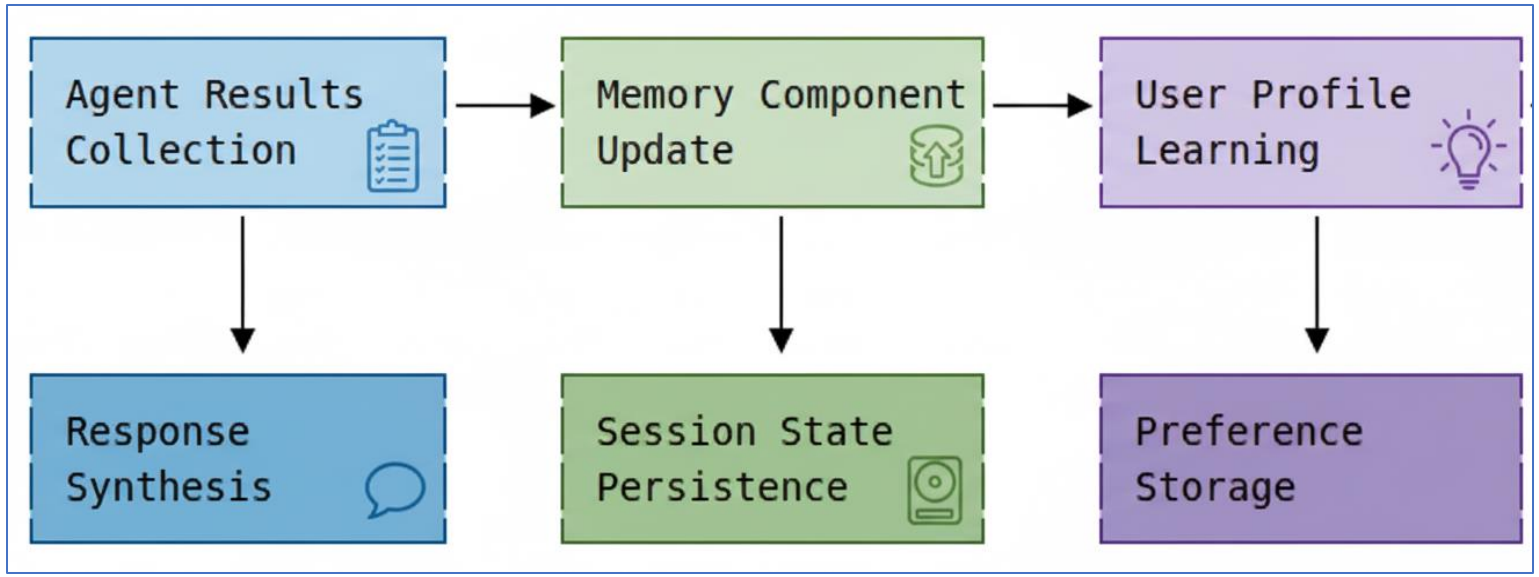
Task Planning and Decomposition

Phase 3

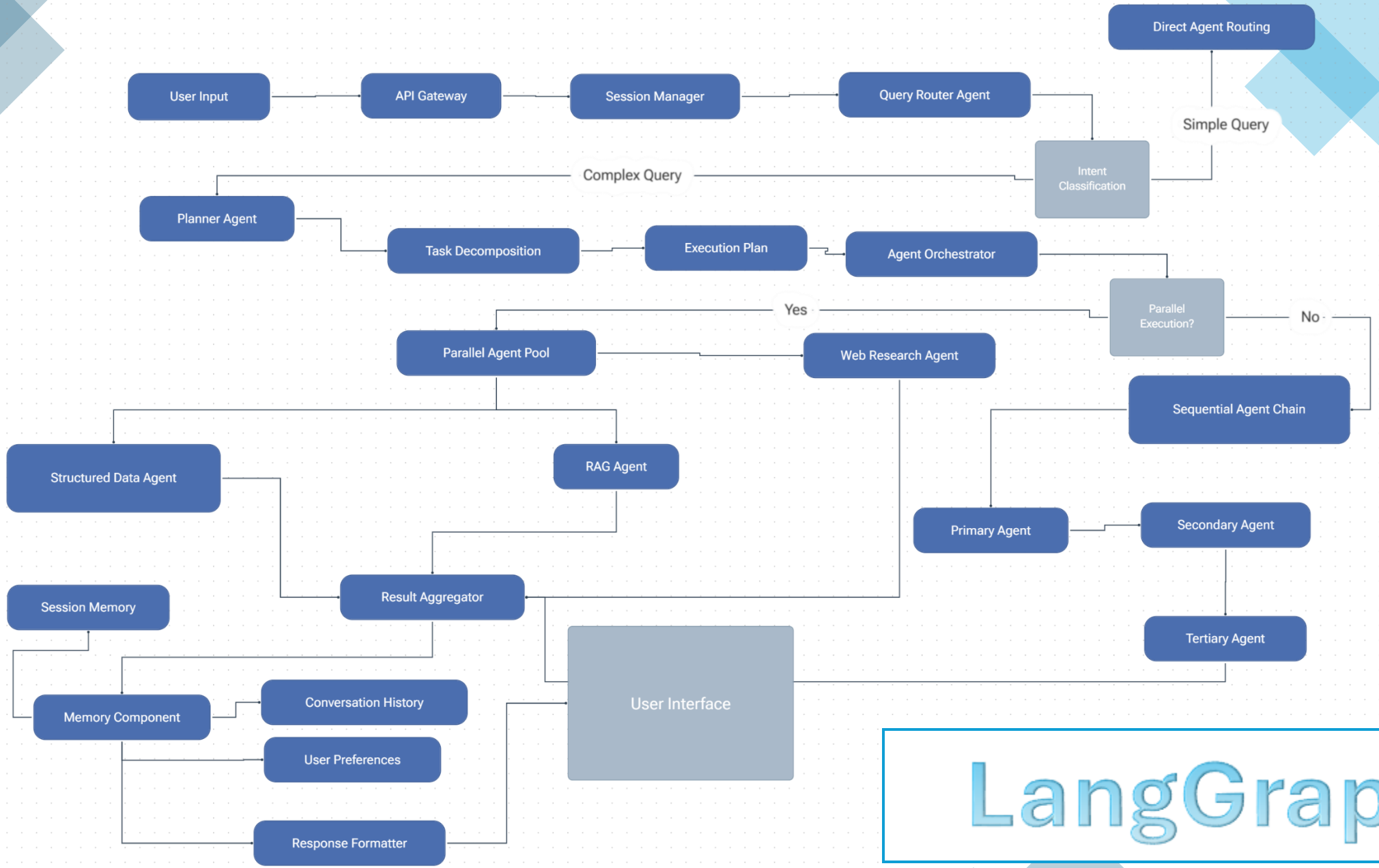


Agent Execution and Communication

Phase 4



Memory Updation and Response Generation


Direct Agent Routing
User Input
API Gateway
Session Manager
Query Router Agent
Simple Query
Complex Query
Intent Classification
Planner Agent
Task Decomposition
Execution Plan
Agent Orchestrator
Parallel Execution?
Yes
No
Parallel Agent Pool
Web Research Agent
Sequential Agent Chain
Structured Data Agent
RAG Agent
Primary Agent
Secondary Agent
Session Memory
Result Aggregator
Tertiary Agent
Memory Component
Conversation History
User Interface
User Preferences
Response Formatter

LangGraph

Key Decisions

Model Selection Decisions

- **Groq over OpenAI** - Chosen for faster inference speeds (2-3x) and cost efficiency for production deployment
- **HuggingFace Sentence Transformers** - Selected for local embedding generation without API dependencies
- **FAISS over ChromaDB** - Opted for Facebook's FAISS for superior performance with large-scale vector similarity search

Technology Stack Decisions

- **FastAPI over Flask** - Selected for async capabilities, automatic API documentation, and production performance
- **Streamlit Frontend** - Rapid prototyping capability with built-in charting and interactive components
- **LangChain Ecosystem** - Unified framework for LLM integration, agent management, and workflow orchestration
- **Redis Memory Backend** - Chosen for session persistence with JSON fallback for development environments

Key Decisions

LangGraph Orchestration Decisions

- **State-Based Workflow** - Chose stateful execution over stateless for complex multi-step property searches
- **Conditional Routing** - Implemented dynamic agent selection based on query complexity and intent analysis
- **Parallel + Sequential Execution** - Hybrid approach allowing independent agents to run parallel while maintaining dependencies
- **Comprehensive Result Synthesis** - Centralized aggregation of multiple agent outputs into coherent responses

Challenges faced

Agent Selection Conflicts - Renovation agent triggered for property searches instead of search agents.

Multi-Agent Coordination - Managing 8 AI agents without conflicts or redundant operations.

Memory Component Errors - Remembering cross session chats made issues

Conda Package Dependencies - System-specific file paths preventing deployment portability.

LangChain Version Conflicts - Incompatible dependency versions causing installation failures.

FAISS Vector Integration - Complex embedding setup breaking semantic search functionality.

State Management Issues - Agent results getting lost during multi-agent workflow executions.

Performance Bottlenecks - 5-10 second response times from simultaneous agent execution.

Web search agent –

unable to fetch the market analysis due to API issues

Improvements

Redis Caching - Implement caching layer for faster query responses and agent results.

Async Agent Execution - Enable true parallel processing for independent agents.

Mobile Application - Develop native iOS/Android apps with offline capabilities.

GPT-4 Integration - Upgrade to advanced language models for better query understanding.

Real-time Market Data - Integrate live property price feeds and market indicators.

ML Recommendation Engine - Build user behavior-based property suggestion system.

Voice Interface - Implement speech-to-text for hands-free property searching.

Advanced Filters - Add granular search options like commute time and school ratings.

Price Prediction AI - Develop time-series forecasting for property value trends.

User Authentication - Implement OAuth login with Google/Facebook integration.

Interactive Maps - Create advanced mapping with property overlays and heatmaps.

WhatsApp Bot - Enable property search through messaging platforms.